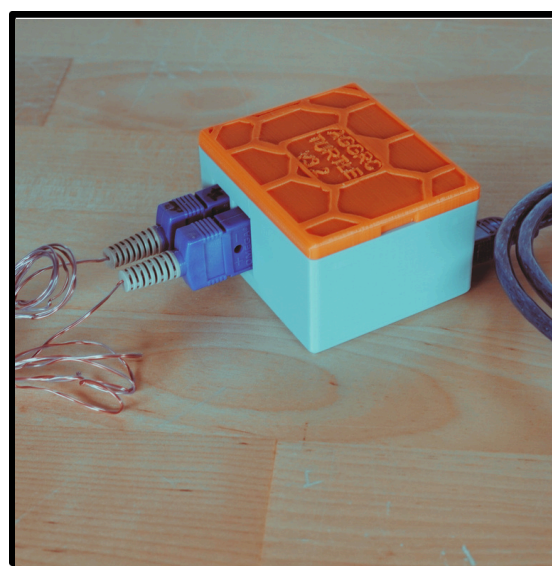


Aquatic Germplasm and Genetic Resources Center	INSTITUTION LSU AgCenter	PROCEDURE ID:
	MANUAL Open Hardware	EFFECTIVE DATE:
	SUBJECT TURTLE V3.6 User Manual	REVISED/REVIEW:

TURTLE V3.6.3 User Manual



Technical Support:

Cole Brumfield, (cbrum16@lsu.edu)

Please send us your comments and suggestions!



*Scan to learn
more about the AGGRC!*



National Institutes of Health
Office of Research Infrastructure Programs



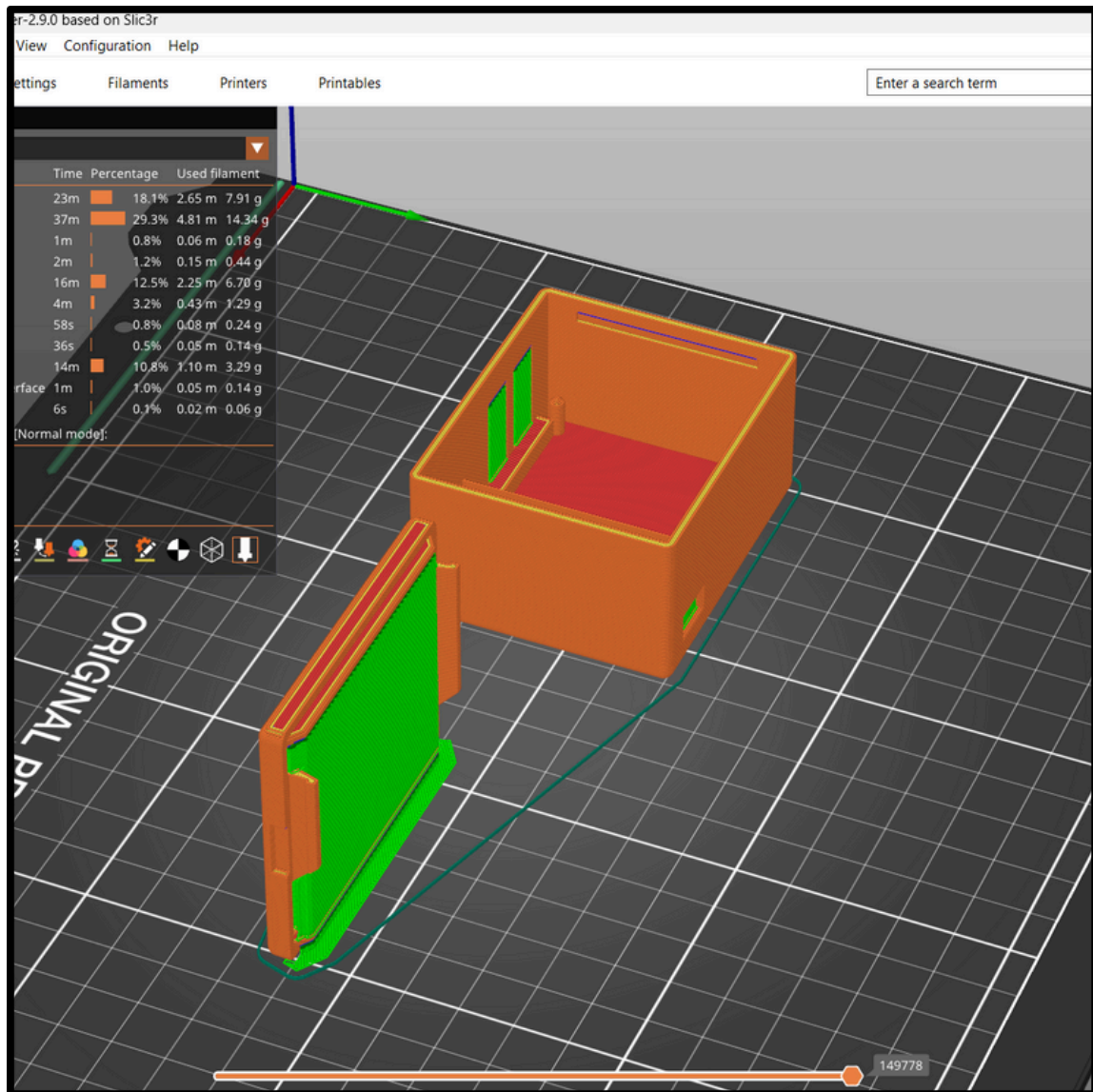
Table of Contents

***Note:** If receiving an assembled TURTLE skip to the TURTLE App Setup section (page 12)

Page #	Content
1	Printing Orientation
2	TURTLE Materials List
3-6	Assembly
7-11	Uploading Arduino Code
12	TURTLE App Setup
13-15	Using TURTLE App



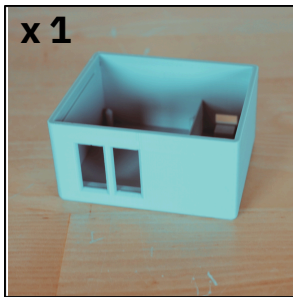
Printing Orientation



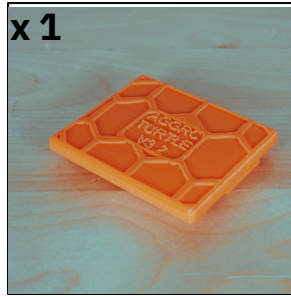
It is recommend to print the case in this orientation. Supports are also necessary (in green)



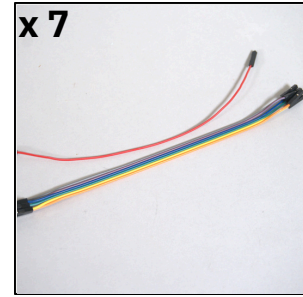
TURTLE Materials



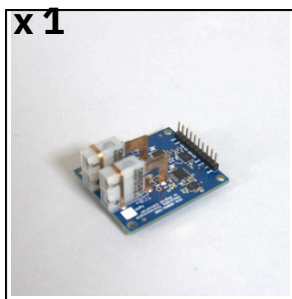
Case (Base)
***Required**



Case (Cover)
***Required**



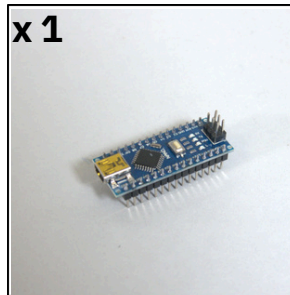
Breadboard
wires(female-female)
***Required**



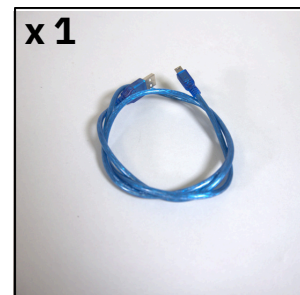
2-Channel
MAX31856
Breakout
***Required**

Link:

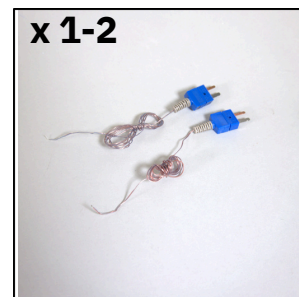
<https://www.playingwithfusion.com/productview.php?pdid=62&catid=1004>



Arduino Nano
(ATmega328P)
***Required**



USB-A to Mini USB
Cable
***Required**



Any type
thermocouple
***Required**

***3-D files can be found at:**
<https://github.com/Colebrumfield/TURTLE-Device>



Assembly

Step 1: Attach Wires between boards

Using breadboard wires make the following connections between the **Thermocouple Board(MAX31856)** and the **Arduino Nano**.

***Note:** Breadboard wire color may vary

Thermocouple Board	Breadboard Wire	Arduino Nano
GND		GND
V+		VIN
CS0		D9
SCK		D13
SDO		D12
SDI		D11
CS1		D10



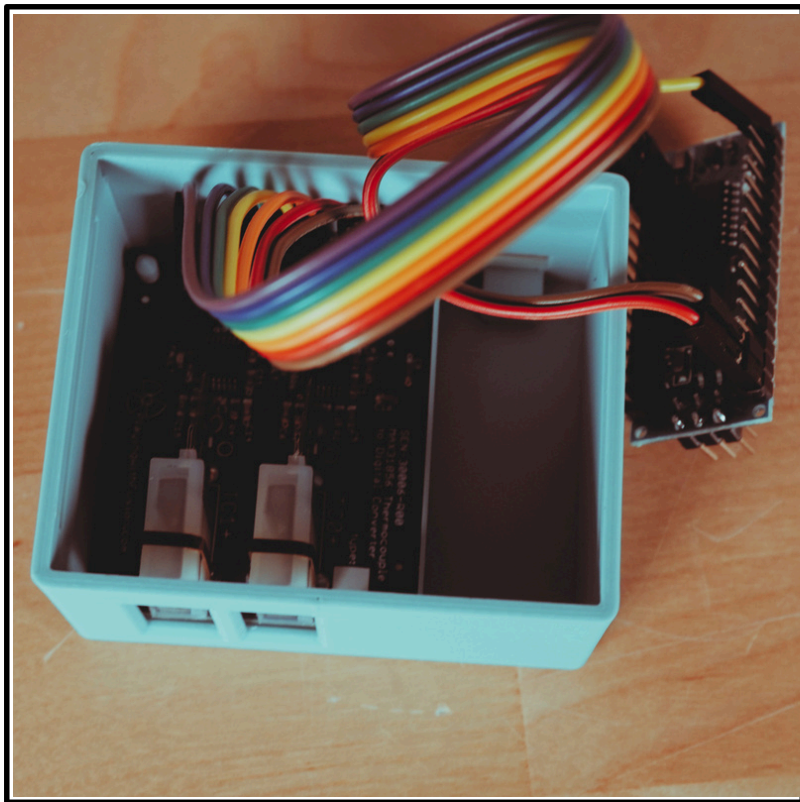
Assembly Continued

Step 2: Orientate the MAX31856 inside of the case

Push the **MAX31856** board into the left side of the case.



Before



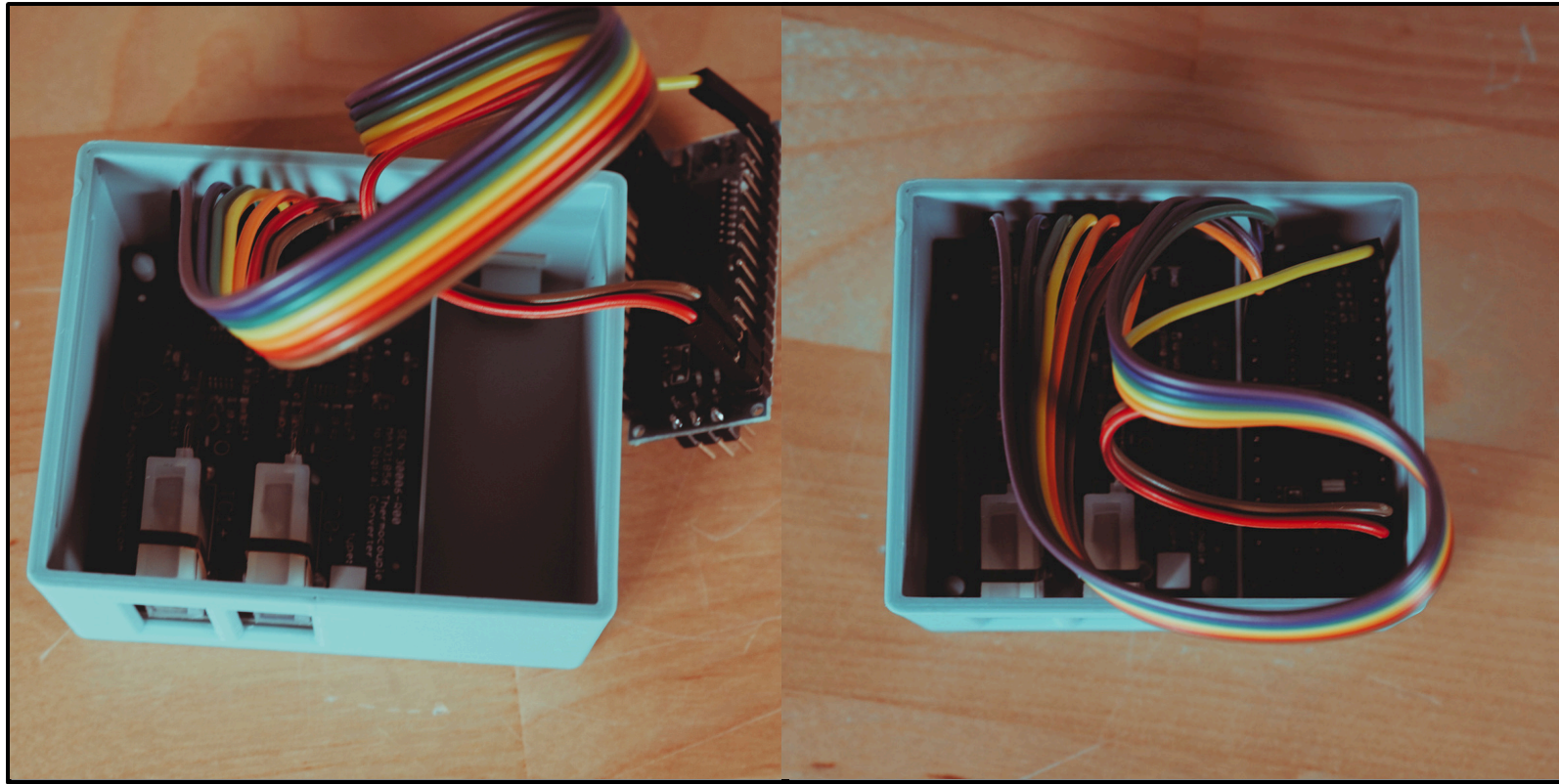
After



Assembly Continued

Step 3: Orientate the Arduino Nano inside of the case

Push the **Arduino Nano** into the right side of the case. Make sure the Arduino Nano is pushed down enough so that the port can be accessed.



Before

After



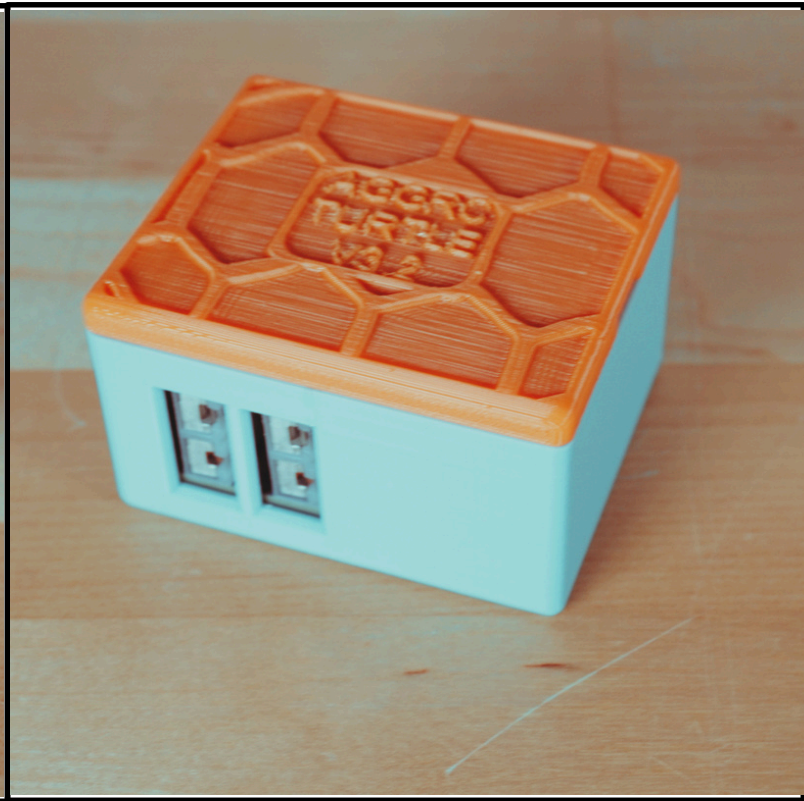
Assembly Continued

Step 4: Snap on cover

Snap the cover onto the case. It is easier to start with one side aligned.



Before



After



Uploading Arduino Code

Step 1: Download the Arduino IDE

Download the Arduino IDE here:

<https://www.arduino.cc/en/software>. Once downloaded, install the program.

Step 2: Download Files

Navigate to the GitHub:

<https://github.com/Colebrumfield/TURTLE-Device/releases/tag/v3.6.3>

From there download:

1. **TURTLE.App.Installer.exe**
2. **Source code (zip)**

Extract both folders

Step 3: Open Arduino Code

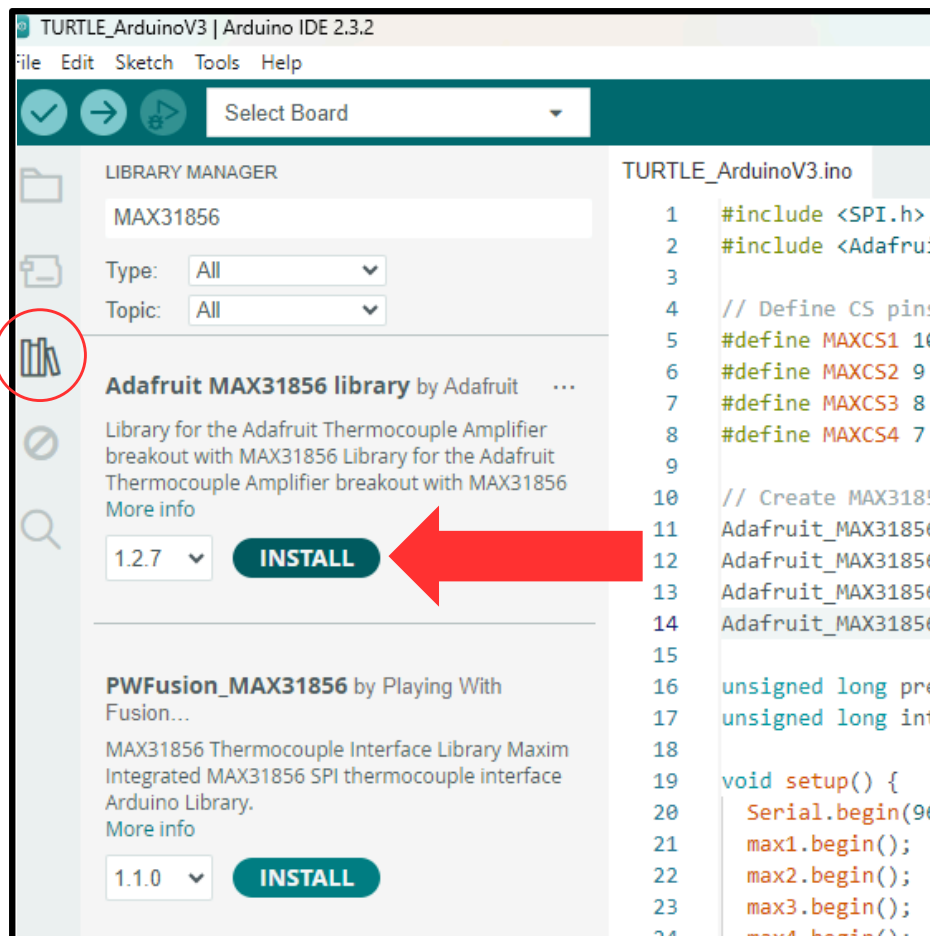
Navigated to the extracted Source code folder. From there open the file named “**TURTLE_ArduinoV3.ino**” using the Arduino IDE



Uploading Arduino Code

Step 4: Install Libraries

Inside the Arduino IDE click on the icon  (that looks like books) on the left. Type in the search bar “**MAX31856**”. Click install under the version titled “**Adafruit MAX31856 library**”.

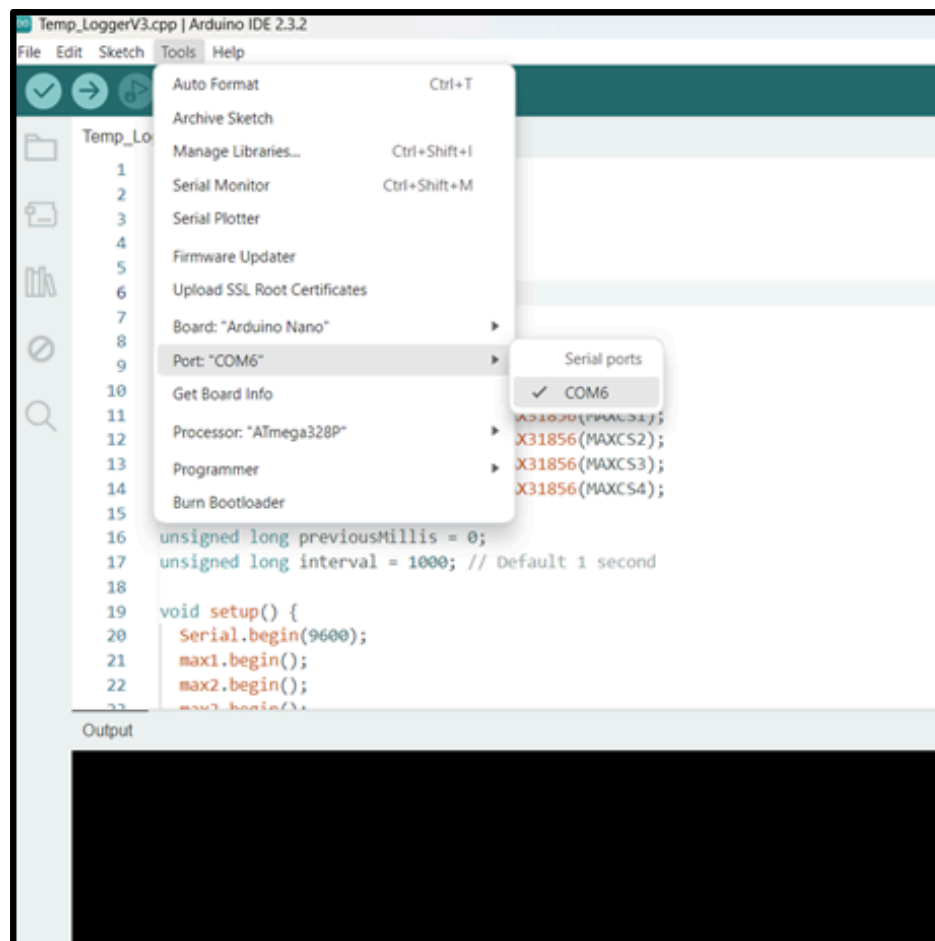


Uploading Arduino Code

Step 5: Selecting port

Plug in the TURTLE Device to your laptop/pc using the USB cable. Inside the Arduino IDE push the “**Tools**” drop down menu. Select “**Port:**”. From there select the port assigned to your Arduino.

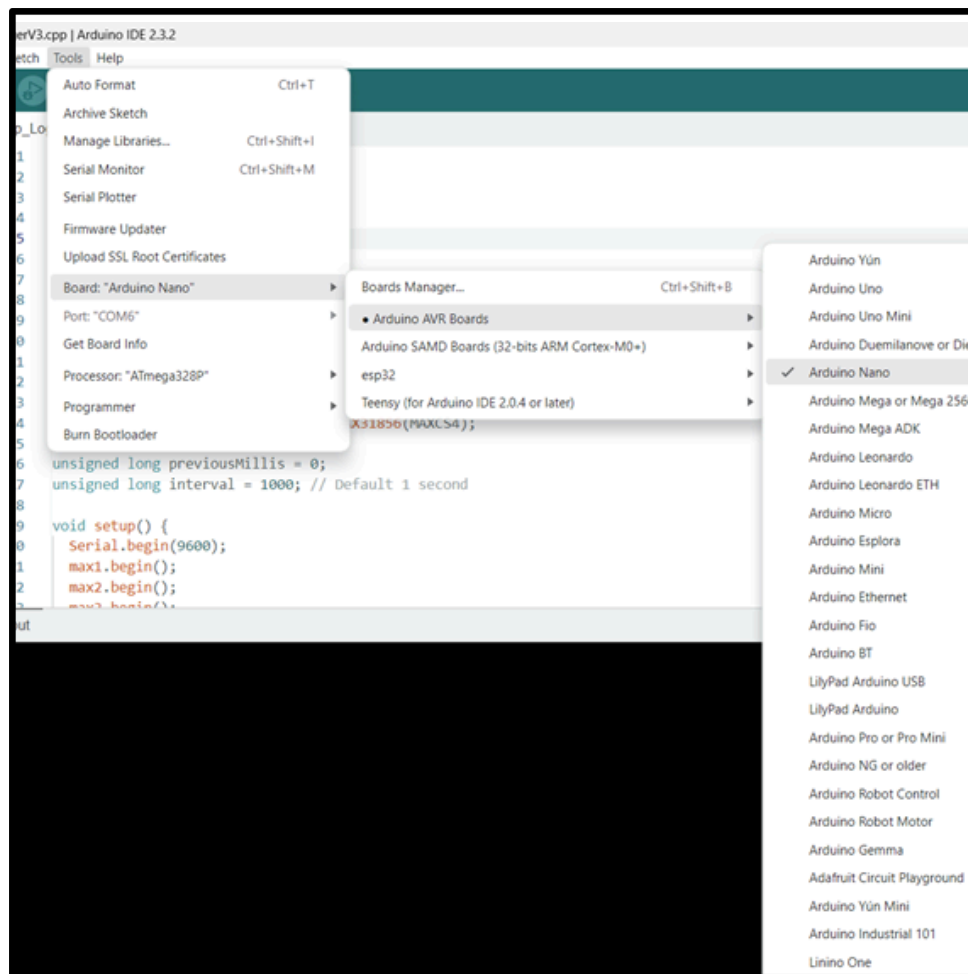
***Note:** Multiple ports may appear (from mice, keyboards, etc.)



Uploading Arduino Code

Step 6: Selecting Board

Push the “**Tools**” drop down menu. From the tools drop down navigate to “**Board:**” and then “**Arduino AVR Boards**” and select “**Arduino Nano**”

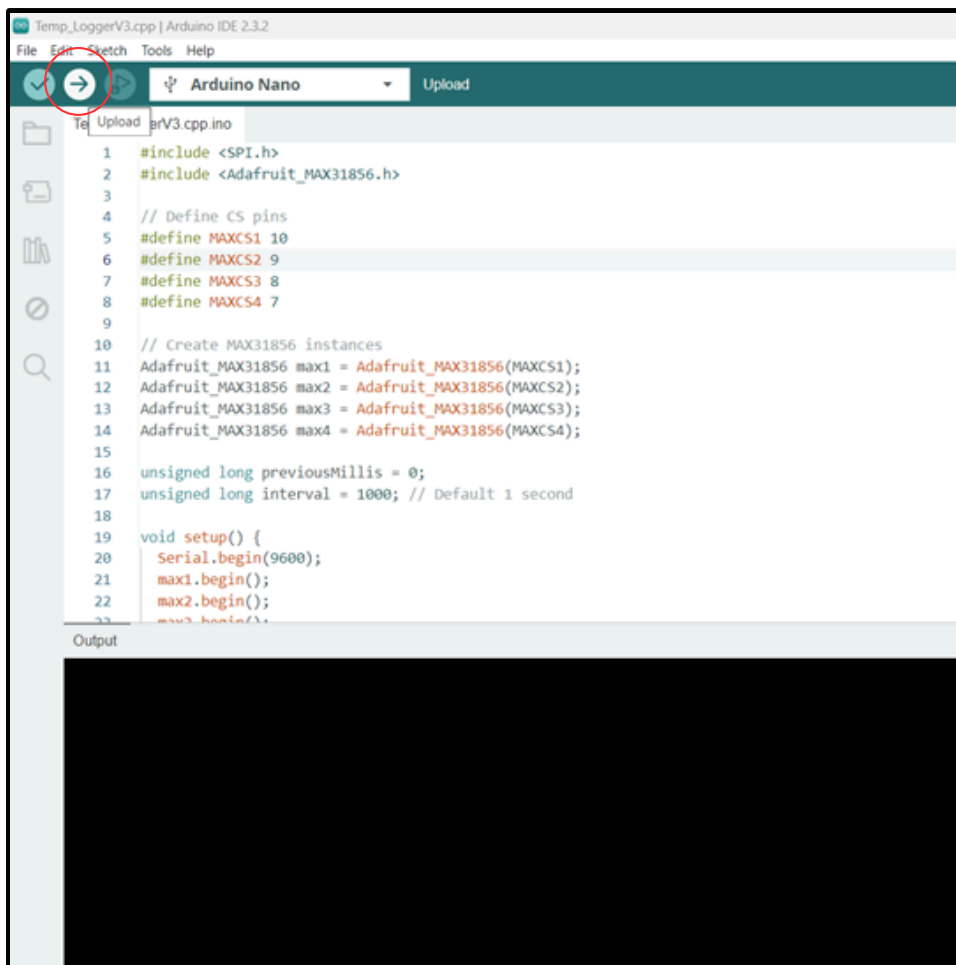


Uploading Arduino Code

Step 7: Uploading Code

Finally push the arrow button at the top left of the screen which will flash the code onto the Arduino.

***Note:** If uploading process takes a long time you may have selected the wrong port. Choose a different port and push the upload arrow again.



TURTLE App Setup

Step 1: Download the Files

Navigate to the GitHub:

<https://github.com/Colebrumfield/TURTLE-Device/releases/tag/v3.6.3>

From there download:

1. **TURTLE.App.Installer.exe**
2. **Source code (zip)**

Extract both folders

Step 2: Download the Driver

Navigated to the extracted Source code folder. From there open the folder named “Windows-CH340-Driver”. Once inside the folder double click the file named “SETUP” to install the driver. This is a driver which will allow your laptop/pc to connect to the TURTLE

Step 3: Run the App

Run **TURTLE.App.Installer.exe**. Make sure to check the “**create desktop shortcut**” check box. This will download the files need for the app as well as a desktop shortcut on your device. Once complete you can delete the installer.

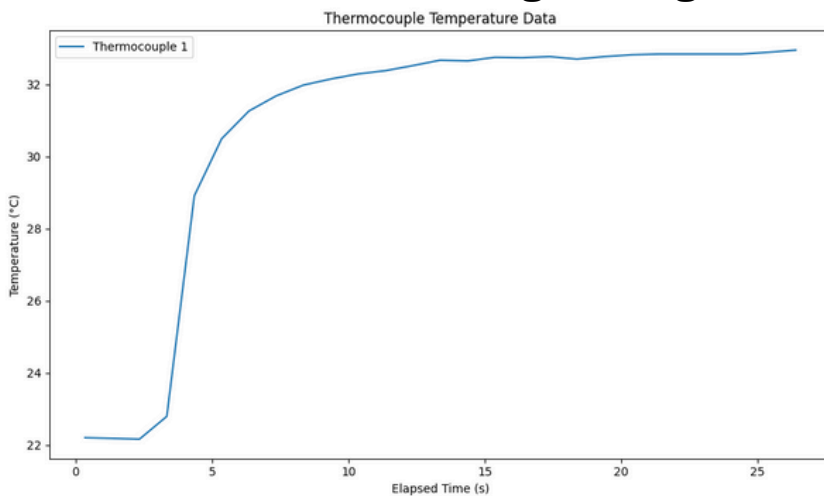


Using TURTLE App

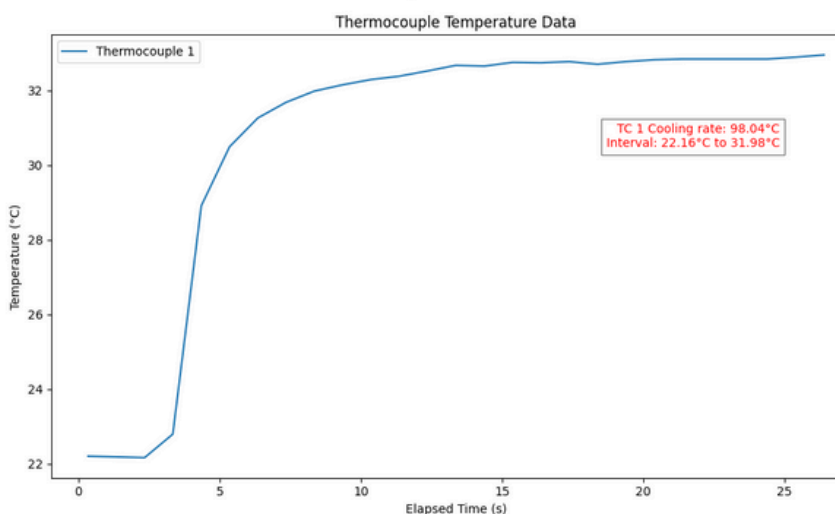
***Note: The TURTLE must be connected to a laptop/pc at all times to function. The TURTLE has no on board memory or power yet.**

Calculating Cooling Rate:

The check box next to “Calculate Cooling Rate” must be checked in order for graphs and excel documents to display cooling rate. Use the start temp and ending temp entry boxes to enter the interval in question. Make sure you are entering the correct temperature unit into these boxes. The resulting cooling rate will be in F or C/min



**Without
“Calculate Cooling Rate”
checked**



**With
“Calculate Cooling Rate”
checked**



Using TURTLE App

Exporting Options:

Data can be exporting either as excel or graph. When saving as a graph push the “Graph Last Data Points” button and then push the save icon in the top left.  When saving as an excel file push the “Export Last Data to Excel” button. Then choose where to save the file.

Problems Installing Driver:

You will need admin access to download this driver. This is a windows driver which is used to communicate with the board inside of the TURTLE. Drivers for different operating systems can be found on the [GitHub](#)

Thermocouple Connection Issues:

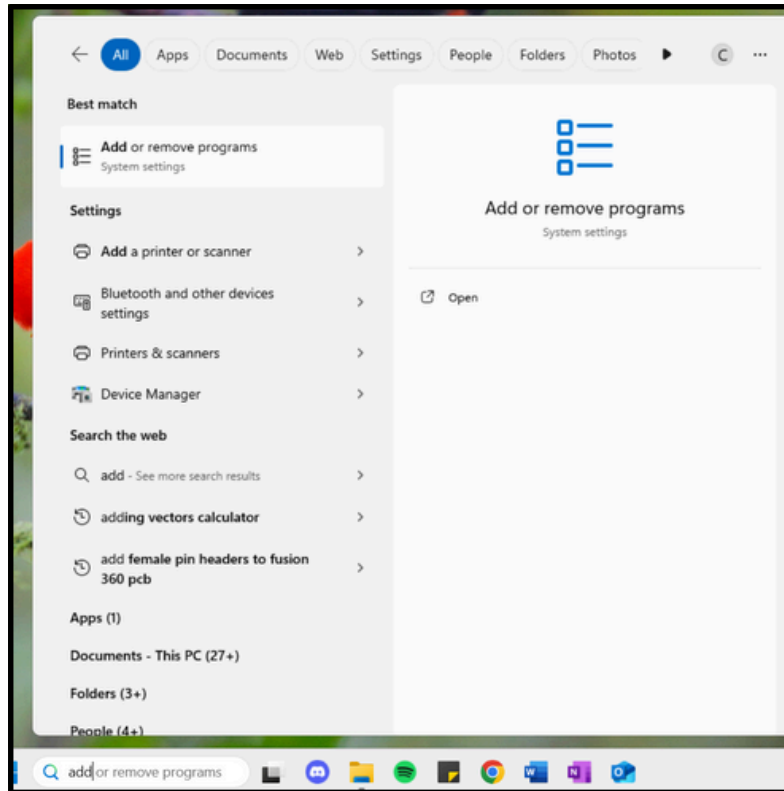
Some thermocouples do not connect to the device as well as others. It is best to plug in the thermocouples into the device and then start the application.



Using TURTLE App

Uninstalling Program:

Search Add or remove programs in the windows search bar



Scroll until you see the TURTLE App and uninstall from there

